

# SARAVANAVEL E

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[in](#) saravanavel-e — [G](#) SaravanavelE — [globe](#) Portfolio

## Profile Summary

Founder of ULockAI , Chairperson in IEEE KCE SB CIS , Undergraduate student in Artificial Intelligence and Data Science with strong foundations in machine learning, data structures, and Python-based development. Experienced in building AI-driven applications and data pipelines. Passionate about applying AI and data-driven techniques to solve real-world problems, published 3 research papers, and active conference technical reviewer.

## Education

### Bachelor of Technology (B.Tech) - Artificial Intelligence and Data Science

Karpagam College of Engineering, Coimbatore.

2023 – 2027

CGPA: 8.05 / 10 (upto Vth Semester)

### Higher Secondary Education - PCBM

Everest Mariappa Nadar Higher Secondary School, Kovilpatti.

2021 – 2023

Percentage: 92%

## Technical Skills

**Programming:** Python, Java

**Data Analysis & Processing:** NumPy, Pandas, Feature Engineering

**Machine Learning:** Supervised Learning, Unsupervised Learning, Model Evaluation

**Deep Learning & NLP:** CNNs, RNNs, Transformers, NLP

**Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-learn, Keras, FastAPI, Flask

**Data Visualization:** Tableau, Power BI

**Databases:** PostgreSQL, Snowflake

**Cloud & Tools:** AWS, GCP, Docker, Git, GitHub, Jupyter Notebook, VS Code

## Soft Skills

Leadership, Collaboration, Problem Solving, Adaptability, Time Management, Communication

## Projects

- **AI-Based Traffic Management System** [G](#) GitHub Mar 2026  
**Tech Stack:** Python, OpenCV, TensorFlow, CNN, YOLO, Flask, CrewAI  
Developed an AI-based traffic management system using YOLO-based vehicle detection and a CrewAI-based multi-agent architecture for adaptive signal control and congestion prediction. Designed agents for traffic analysis, prediction, signal optimization, emergency prioritization, and feedback-driven learning.
- **Data Analytics Model to Reduce Rejection Rates in Small Industries** [G](#) GitHub Feb 2026  
**Tech Stack:** Pandas, NumPy, Matplotlib, Seaborn, Linear Regression, Streamlit  
Developed a data analytics system applying Pareto Analysis and Statistical Process Control (SPC) to identify production bottlenecks and reduce rejection rates in small-scale manufacturing industries.
- **Real-Time Cryptocurrency Streaming with Kafka & Snowflake** [G](#) GitHub Jan 2026  
**Tech Stack:** Python, Apache Kafka, CoinGecko API, Snowflake, Docker  
Built a real-time data pipeline to ingest live cryptocurrency prices via CoinGecko API, stream data via Apache Kafka, and store structured data in Snowflake for analytics and visualization.
- **Eye Blink Detection Using Deep Learning for Eye Safety** [G](#) GitHub Oct 2025  
**Tech Stack:** Python, TensorFlow, Keras, CNN, MobileNetV2, OpenCV, MediaPipe, Flask  
Developed a deep learning-based eye blink detection system using CNN models and MediaPipe facial landmarks to analyze blink patterns and detect eye fatigue.

## Languages

English (Professional), Tamil (Native), Hindi, Japanese (Elementary)

Internships

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>InternPro – Artificial Intelligence Intern</b><br>Developed AI agents for weather forecasting using Python and machine learning models.                       | <i>Jul 2025 – Sep 2025</i> |
| <b>Techno Hacks Solutions Pvt. Ltd. – Data Science Intern</b><br>Applied data analysis and ML techniques on real-world datasets; created interactive dashboards. | <i>Sep 2024 – Oct 2024</i> |

Hackathons & Competitions

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Smartathon 2.0 Hackathon (Team: Zentrix)</b><br><b>Top 10 Finalist</b> — Conducted by <b>L&amp;T</b> and IEEE IAS at St. Joseph College of Engineering, Chennai<br>Project: AI-Based Traffic Management System | Mar 2026            |
| <b>TN Impact Hackathon (Team: Algniters)</b><br>Conducted by <b>TN Govt.</b> at Sona Institute of Technology, Salem<br>Project: Data Analytics Model to Reduce Rejection Rates in Small Industries                | Feb 2026            |
| <b>4th Hack Nation Hackathon (Team: Algniters)</b><br>Conducted by <b>OpenAI, MIT</b><br>Project: Universal Web Agent                                                                                             | Feb 2026            |
| <b>HACKCELR8 (Team: Deep Seekers)</b><br><b>Advanced to Round 2</b> , Conducted by IEEE CS at SRMIST, Chennai<br>Project: Built an AI-based loyalty and recommendation platform                                   | Sep 2025            |
| <b>FIFS Gameathon 2.0 (Team: CricTech)</b><br><b>13th Place Nationwide</b> — Conducted by FIFS<br>Project: AI model for Dream11 cricket team prediction                                                           | Mar 2025 – Jun 2025 |

Certifications

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|------------------------------------------------------------|----------|
| <b>Oracle Generative AI Professional</b> – Oracle (OCI)    | Oct 2025 |
| <b>Oracle AI Vector Search Professional</b> – Oracle (OCI) | Oct 2025 |
| <b>AI Agents Intensive Course</b> – Google & Kaggle        | Dec 2025 |
| <b>Advanced Deep Learning with Keras</b> – DataCamp        | Aug 2025 |
| <b>Transformer Models with PyTorch</b> – DataCamp          | Aug 2025 |
| <b>Python for Data Science</b> – NPTEL (Elite + Silver)    | Dec 2024 |

Publications

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| <b>Deep Learning–Driven Solar Power Forecasting for Smart Grid Optimization</b><br>Accepted and presented at the <i>Fourth International Conference on Electronics and Renewable Systems (ICEARS 2026)</i><br>DOI: 10.1109/ICEARS67481.2026.11416716 | Feb 2026 |
| <b>AI-powered Sustainable Smart Agriculture using UAV and IoT Metadata</b><br>Published in the <i>10th International Conference on Communication and Electronics Systems (ICCES 2025)</i> .<br>DOI: 10.1109/ICCES67310.2025.11336861                 | Oct 2025 |

Reviewer Experience

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| <b>Technical Reviewer</b> – International Conference on AI and Security for Industrial IoT Systems ( <b>AI-SIIS 2026</b> ) | Mar 2026 |
| <b>Technical Reviewer</b> – Global Symposium on Emerging and Communication Technologies ( <b>GSEACT 2026</b> )             | Mar 2026 |
| <b>Technical Reviewer</b> – 4th International Conference on Computational Intelligence & Systems ( <b>CCIS 2026</b> )      | Feb 2026 |

Leadership & Volunteering

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|------------------------------------------------------------------------------------|---------------------|
| <b>Chairperson</b> , IEEE KCE SB CIS, Karpagam College of Engineering, Coimbatore. | Aug 2025 – Present  |
| <b>Secretary</b> , IEEE KCE SB CIS, Karpagam College of Engineering, Coimbatore.   | Oct 2024 – Jul 2025 |

Professional Affiliations

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| <b>International Association of Engineers (IAENG)</b> – Student Member           | Oct 2025 – Present |
| <b>Institute of Electrical and Electronics Engineers (IEEE)</b> – Student Member | Oct 2024 – Present |